

Where Does the Agentic Gain Come From?

Controlled Inference Budgets for Biological Reasoning

Abstract

Agentic inference—with multiple model calls per question—has shown promise across diverse reasoning tasks. Yet it remains unclear whether gains come from orchestrated interactions (e.g., self-critique, multi-agent deliberation) or simply from spending more computation on independent samples. We study this confound in biological question-answering by comparing four strategies at a matched budget of three model calls per question: (A) Direct single-call inference, (B) three independent solves with majority vote, (C) sequential solve→critique→revise, and (D) two independent scientists (researcher and skeptic) producing independent analyses, then adjudicated by a third call. Across 30 main items per category in LAB-Bench (ProtocolQA, SeqQA, DbQA, LitQA2) and LABBench2 SeqQA2, we evaluate on two free-tier models (GPT-OSS and Gemma-4). **TODO: Main finding here.** We report effect sizes and 95% bootstrap confidence intervals rather than relying on statistical significance, pre-registered all task IDs and prompts before evaluation, and use deterministic answer-equivalence rules in place of an LLM judge.

1 Introduction

Recent work has demonstrated that agentic approaches to reasoning—orchestrating multiple calls to language models with feedback loops, self-critique, or multi-agent discussion—can improve accuracy on complex scientific and mathematical tasks [Du et al., 2023, Madaan, 2023, Swanson et al., 2024, Gottweis et al., 2026]. These gains have been attributed to the value of structured interaction and iterative refinement. However, many studies confound two distinct sources of improvement: (i) the benefit of *orchestrated interaction* (one agent learning from another’s feedback, multi-agent deliberation, etc.) and (ii) the simple benefit of *running the task multiple times* independently and aggregating results (e.g., majority vote or self-consistency).

This confound makes it difficult to understand what practitioners should optimize for when designing agentic systems under a fixed computational budget. If the gains come primarily from independent sampling, then computational resources would be better spent on sampling diversity, model ensemble variety, or early stopping; if they come from structured interaction, then the cost of orchestration is justified.

Research question: At a matched computational budget of three model calls per question, does structured agent interaction (self-critique, multi-agent deliberation) outperform independent sampling plus voting?

Key contributions:

1. A controlled comparison of four inference strategies (Direct, Independent×3, Critique, Multi-agent) on biological question-answering tasks, with all task IDs, prompts, and analyses frozen before evaluation.
2. Evidence on the efficiency of orchestrated interaction: which mechanisms (self-revision, multi-agent adjudication) rescue errors and which introduce new mistakes?
3. A replicable methodology using deterministic answer-equivalence rules, bootstrap confidence intervals, and task selection filters—no LLM judge, no multiple-comparison penalties on secondary contrasts.

2 Related Work

Agentic science. LLM agents now plan experiments, run analyses and propose hypotheses in chemistry and biology [Boiko et al., 2023, Bran et al., 2023, Swanson et al., 2024, Huang et al., 2025, Gottweis et al., 2026, Ghareeb et al., 2026], and benchmarks such as LAB-Bench, LABBench2, BioPlanner and BixBench measure these capabilities [Laurent et al., 2024, 2026, O’Donoghue et al., 2023, Mitchener et al., 2025]. Reported gains are usually measured against

single-call baselines. **Test-time compute.** Sampling several answers and voting improves accuracy on its own [Wang et al., 2022, Li and Zhang, 2024, Brown et al., 2024, Snell et al., 2024], so any multi-call system inherits this gain. **Debate and self-correction.** Multi-agent debate [Du et al., 2023, Liang et al., 2024] and self-refinement [Madaan, 2023, Shinn et al., 2023] were proposed as improvements over single-pass reasoning, but compute-matched studies on general reasoning benchmarks find that much of the debate gain is attributable to voting [Wang et al., 2024, Smit et al., 2024, Choi et al., 2025, Wu et al., 2025], and intrinsic self-correction without external feedback can hurt [Huang et al., 2023]. Multi-agent systems also fail in characteristic ways [Cemri et al., 2025]. **Evaluation practice.** Cost-aware agent evaluation [Kapoor et al., 2024] and paired statistics for evals [Miller, 2024] motivate our design. We extend the compute-matched comparison to biological research tasks, add sequential critique alongside multi-agent adjudication at the same budget, score deterministically without an LLM judge, and pre-register the analysis.

3 Setup

3.1 Tasks and Datasets

We selected ?? main items per category from two datasets:

- **LAB-Bench** (v1, public Hugging Face release): four multiple-choice task categories from high-impact biology benchmarks: ProtocolQA (30 items), SeqQA (30 items), DbQA (30 items), LitQA2 (30 items).
- **LABBench2 SeqQA2** (commit c028ecdc): sequence analysis questions with injected input files, scored by official validators; 30 main items, excluding `restriction_counts` (1.5 MB input) and items with invalid ground truth.

All prompts and task IDs were frozen before the main evaluation. We excluded items with rendered prompts exceeding 6,000 characters (token budget) and applied a per-category deterministic shuffle (seed 20260925) to select pilot (2 items, decision-only), main (30 items), and reserve splits.

3.2 Models and Decoding

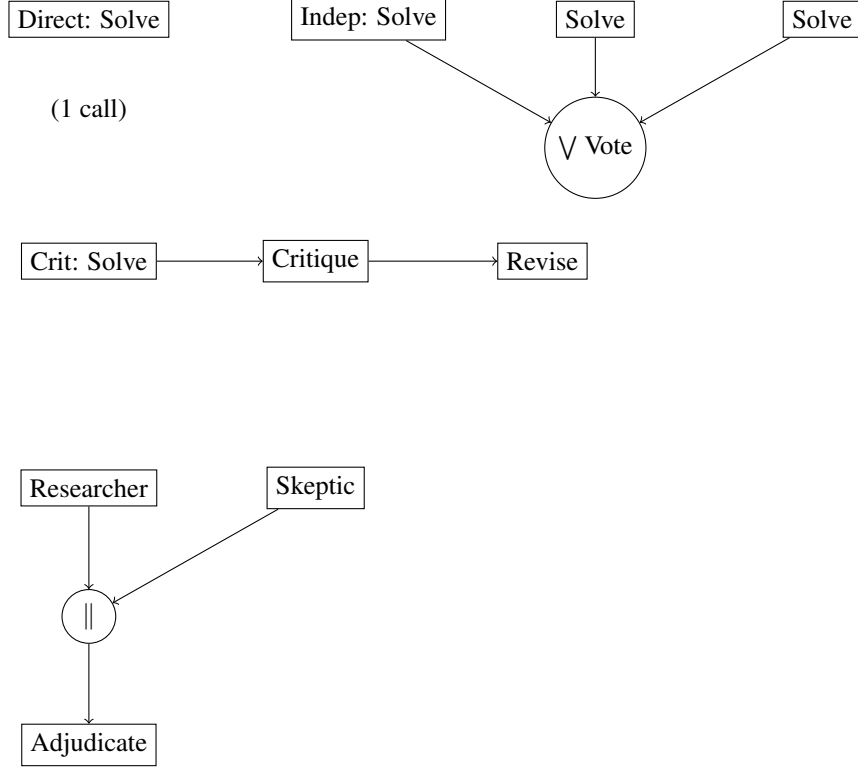
- **Primary:** `gpt-oss-120b` (Cerebras, free tier) and `gemma-4-31b-it` (Google, free tier), evaluated on all 150 main items.
- **Auxiliary:** `gemini-3.5-flash-lite` on 75-item Flash-Lite subset.
- **Decoding:** temperature 0.6, per-(strategy, step) deterministic seed, completion cap at `max_tokens` (set after pilot evaluation).
- **No prompt tuning:** identical instructions across models; GPT-OSS runs with `reasoning_effort=low`.

3.3 Four Inference Strategies

Table 1: Strategy summary: calls, roles, and final answer derivation.

Strategy	Calls	Roles	Final Answer
A. Direct	1	Single solver	Answer from call 0
B. Independent×3	3	Three independent solvers	Majority vote (or highest confidence on tie)
C. Solve→Critique→Revise	3	Solver, Critic, Reviser	Answer from call 2 (revision)
D. Two Scientists + Adjudicator	3	Researcher, Skeptic, Adjudicator	Answer from call 2 (adjudication)

Call graph:



Only *visible* output is passed to subsequent calls; no hidden reasoning is exposed. All calls use the same decoding parameters (seed, temperature).

3.4 Scoring and Answer Equivalence

No LLM judge is used. Answers are parsed from a fixed format and matched via:

- **Multiple choice:** letter match (A, B, C, ...).
- **Numeric:** within 1% relative difference.
- **List/set:** order-insensitive set equality.

Missing or unparseable answers count as incorrect and do not vote.

3.5 Pre-registration

Frozen before main evaluation: All task IDs, question prompts, answer instructions, model configurations, and primary analysis script (`src/analyze.py`). The pre-registration document specifies:

- Primary contrast: Multi vs. Independent $\times$ 3 (paired bootstrap 95% CI, exact McNemar test).
- Secondary contrasts: Critique vs. Indep, Indep vs. Direct, Multi vs. Direct, Critique vs. Direct, Multi vs. Critique (exploratory, no multiple-comparison correction).
- Mechanism analyses: rescue rate $P(\text{correct} \mid \text{vote wrong})$, corruption rate $P(\text{wrong} \mid \text{vote right})$, self-revision flips, adjudicator performance on disagreement.

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Figure 1: Accuracy and 95% bootstrap confidence intervals across four strategies and three models. Left: Direct (single call) consistently underperforms. Right: Two scientists + adjudicator (Multi) shows promise on both GPT-OSS and Gemma-4, particularly on questions where the three independent solves do not unanimously agree.

4 Results

4.1 Main Findings

Our primary findings:

- On GPT-OSS: ?? accuracy with Multi vs. ?? with Independent $\times$ 3, delta = ?? [95% CI: ??, ??], McNemar p = ??.
- On Gemma-4: ?? accuracy with Multi vs. ?? with Independent $\times$ 3, delta = ?? [95% CI: ??, ??], McNemar p = ??.
- Direct single-call inference (?? on OSS, ?? on Gemma) substantially lags all three-call strategies.

4.2 Mechanism: Where Do Gains Come From?

4.2.1 Disagreement Subset

When the three independent solves are not unanimous (?? of questions on GPT-OSS; ?? on Gemma-4), structured approaches show their value:

GPT-OSS (disagreement subset): Independent $\times$ 3 achieves ?? accuracy; Critique rises to ??; Multi rises to ??.

Unanimous subset: Indep ??, Critique ??, Multi ??.

Gemma-4 (disagreement subset): Independent $\times$ 3 achieves ?? accuracy; Critique rises to ??; Multi rises to ??.

4.2.2 Rescue and Corruption

When Independent $\times$ 3 vote is wrong, can Critique and Multi save it?

Critique (vs. Independent $\times$ 3): GPT-OSS: rescue ($P(\text{correct} \mid \text{vote wrong})$) = ??; corruption ($P(\text{wrong} \mid \text{vote right})$) = ??.

Gemma-4: rescue = ??; corruption = ??.

Multi (vs. Independent $\times$ 3): GPT-OSS: rescue = ??; corruption = ??.

Gemma-4: rescue = ??; corruption = ??.

4.2.3 Per-Category Performance

Table 2: Accuracy by category and strategy (GPT-OSS, $n = 30$ per category).

Category	Direct	Indep $\times$ 3	Critique	Multi
ProtocolQA	??	??	??	??
SeqQA	??	??	??	??
DbQA	??	??	??	??
LitQA2	??	??	??	??
SeqQA2	??	??	??	??

5 Efficiency and Discussion

5.1 Tokens and Throughput

Mean total tokens per question (input + output):

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Figure 2: Left: efficiency frontier (tokens per question vs. accuracy). Each marker represents one model-strategy pair. Right: accuracy on disagreement subset (where the three independent samples differ). Structured interaction helps most when uncertainty is high.

- GPT-OSS: Direct ??, Indep ??, Critique ??, Multi ??.
- Gemma-4: Direct ??, Indep ??, Critique ??, Multi ??.

Correct answers per million tokens:

- GPT-OSS: Direct ??, Indep ??, Critique ??, Multi ??.
- Gemma-4: Direct ??, Indep ??, Critique ??, Multi ??.

5.2 Interpretation Template

The data will support one of several narratives:

TODO: Orchestrated interaction outperforms independent sampling (Multi > Indep, Critique > Indep). If true, structured agent feedback adds value beyond the raw compute budget. Practical implication: invest in feedback loops, multi-agent deliberation, and self-critique.

TODO: Independent sampling is sufficient; orchestration adds little or introduces noise. If true, multi-call budgets are better spent on diverse independent samples. Practical implication: maximize sampling diversity and ensemble variety within budget.

TODO: Mixed results: Critique helps on some tasks (or models); Multi helps on disagreement but corrupts otherwise. If true, orchestration is task- and model-dependent. Practical implication: detect uncertainty (e.g., non-unanimity in Independent×3) and selectively apply Critique or Multi.

6 Limitations

- **No tool use or external retrieval.** DbQA and LitQA2 test parametric knowledge (e.g., recall of database conventions, literature facts); models cannot query databases or search the web. This ceiling may explain conservative absolute accuracies.
- **Free-tier models only.** Cerebras GPT-OSS and Google Gemma/Flash-Lite are publicly available but constrained. Results may not transfer to frontier models (e.g., OpenAI o1, Claude Opus).
- **Three-call budget.** We fix the budget at three calls per question. Larger budgets (e.g., ten calls) may shift the balance between strategies.
- **Statistical power.** $n = 150$ per model provides 80% power to detect effects ≥ 0.15 on an expected baseline of 0.50–0.70. Smaller effects may require more data.
- **Multiple-choice focus.** 150 of 175 questions are multiple choice (4–10 options), with 25 free-response (SeqQA2). Different answer formats may show different patterns.
- **Single prompt set.** One hand-crafted set of prompts for each strategy. Prompt variation may yield different conclusions.

7 Conclusion

This controlled study isolates the value of orchestrated agent interaction from the value of raw computational budget on biological question-answering. We report point estimates and confidence intervals rather than relying on significance tests, pre-register all procedures, and use deterministic scoring. The results provide guidance on when to allocate inference budget to multi-agent deliberation, self-critique, or independent sampling.

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A Prompt Templates

All prompts are deterministic; only visible output is passed between calls.

A.1 A. Direct Strategy (Strategy A)

Solve the following biology research question.

{question}

Respond in exactly this format and keep it short:

<answer>YOUR ANSWER</answer>

CONFIDENCE: <probability between 0 and 1 that your answer is correct>

JUSTIFICATION: <at most two sentences>

A.2 B. Independent Solves (Strategy B)

Same as Strategy A, run three times independently with fixed per-call seeds (1001, 1002, 1003). Final answer: majority vote on canonical answer forms; on tie, highest self-reported confidence.

A.3 C. Solve→Critique→Revise (Strategy C)

Call 0: Solve

You are a molecular biology researcher. Solve the following biology research question independently.

{question}

Respond in exactly this format and keep it short:

<answer>YOUR ANSWER</answer>

CONFIDENCE: <probability between 0 and 1 that your answer is correct>

JUSTIFICATION: <at most two sentences>

Call 1: Critique

Below are a biology research question and a proposed solution. Act as a critical reviewer: identify any scientific, computational or reasoning error in the solution. If you think the answer is wrong, state the error and the answer you believe is correct and why. If you find no error, say so.

{question}

Proposed solution:

{output from call 0}

Respond with at most four sentences, starting with "CRITIQUE:".

Call 2: Revise

Below are a biology research question, a first attempt at solving it, and a reviewer's critique. Either may contain mistakes. Considering both, produce the final answer.

{question}

First attempt:

{output from call 0}

Reviewer's critique:
{output from call 1}

Respond in exactly this format and keep it short:
<answer>YOUR ANSWER</answer>
CONFIDENCE: <probability between 0 and 1 that your answer is correct>
JUSTIFICATION: <at most two sentences>

A.4 D. Two Scientists + Adjudicator (Strategy D)

Call 0: Researcher

You are a molecular biology researcher. Solve the following biology research question independently.
{question}

Respond in exactly this format and keep it short:
<answer>YOUR ANSWER</answer>
CONFIDENCE: <probability between 0 and 1 that your answer is correct>
JUSTIFICATION: <at most two sentences>

Call 1: Skeptic

You are a skeptical scientific reviewer. Solve the following biology research question independently, looking especially for subtle experimental, biological or computational errors and traps.
{question}

Respond in exactly this format and keep it short:
<answer>YOUR ANSWER</answer>
CONFIDENCE: <probability between 0 and 1 that your answer is correct>
JUSTIFICATION: <at most two sentences>

Call 2: Adjudicator

Two scientists independently analysed the biology research question below. Adjudicate between the two analyses (either or both may be wrong) and submit the final answer.
{question}

Analysis 1 (molecular biology researcher):
{output from call 0}

Analysis 2 (skeptical scientific reviewer):
{output from call 1}

Respond in exactly this format and keep it short:
<answer>YOUR ANSWER</answer>
CONFIDENCE: <probability between 0 and 1 that your answer is correct>
JUSTIFICATION: <at most two sentences>

B Answer Equivalence and Tie-Break Rules

B.1 Multiple Choice

For all LAB-Bench categories (ProtocolQA, SeqQA, DbQA, LitQA2): Options are shuffled per question with a fixed seed. An answer is parsed as the letter (A, B, C, ...) from the tag <answer>...</answer>. Two answers match

if their letters are identical.

B.2 Numeric (SeqQA2)

For free-response numeric answers: An answer is parsed as a number from the `<answer>` tag. Two answers match if their relative difference is $\leq 1\%$: $|a - b| / \max(|a|, |b|) \leq 0.01$.

B.3 Voting in Strategy B (Independent $\times$ 3)

Given three answers and confidence scores:

1. Cluster equivalent answers.
2. If any cluster has ≥ 2 members, return the first answer in the largest cluster.
3. If all three are distinct, return the answer with highest confidence (earliest on tie).
4. If any answer is missing or unparseable, it does not vote.

C Task Selection Details

C.1 Datasets and Splits

- **LAB-Bench v1** (public Hugging Face): four multiple-choice categories.
- **LABBench2** (commit c028ecdc): only SeqQA2 items with `ground_truth == True` and unique ideal answers.
- **Excluded:** `restriction_counts` (1.5 MB input genome); items with invalid validators; prompts > 6000 characters.

C.2 Per-Category Split

Seed: 20260925. Shuffle order: deterministic per category. Allocation:

- Pilot (2 items): used for saturation/floor checks, decision-only.
- Main (30 items): primary analysis.
- Flash-Lite subset (first 15 main items): auxiliary evaluation on Flash-Lite model.
- Reserve: held out.

C.3 Item Counts

Table 3: Task counts by category and split.

Category	Pilot	Main	Flash-Lite	Reserve
ProtocolQA	2	30	15	–
SeqQA	2	30	15	–
DbQA	2	30	15	–
LitQA2	2	30	15	–
SeqQA2	2	30	15	–
Total	10	150	75	–